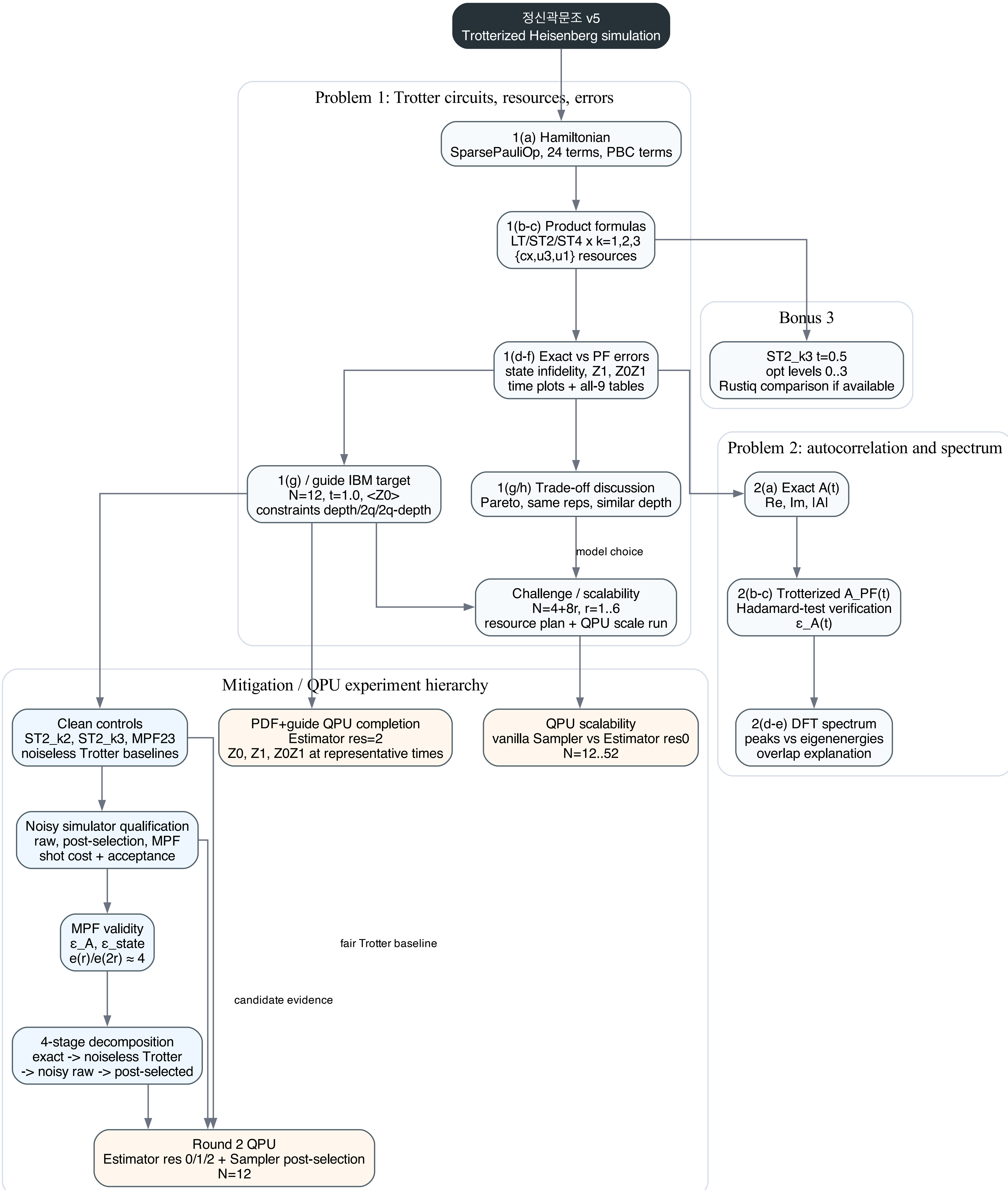

정신곽문조 v5
Trotterized Heisenberg simulation
Problem 1: Trotter circuits, resources, errors
1(a) Hamiltonian
SparsePauliOp, 24 terms, PBC terms
1(b-c) Product formulas
LT/ST2/ST4 x k=1,2,3
{cx,u3,u1} resources
Bonus 3
ST2_k3 t=0.5
opt levels 0..3
Rustiq comparison if available
1(d-f) Exact vs PF errors
state infidelity, Z1, Z0Z1
time plots + all-9 tables
Problem 2: autocorrelation and spectrum
1(g) / guide IBM target
N=12, t=1.0, <Z0>
constraints depth/2q/2q-depth
1(g/h) Trade-off discussion
Pareto, same reps, similar depth
2(a) Exact A(t)
Re, Im, |A|
model choice
Challenge / scalability
N=4+8r, r=1..6
resource plan + QPU scale run
2(b-c) Trotterized A_PF(t)
Hadamard-test verification
ε_A(t)
Mitigation / QPU experiment hierarchy
Clean controls
ST2_k2, ST2_k3, MPF23
noiseless Trotter baselines
PDF+guide QPU completion
Estimator res=2
Z0, Z1, Z0Z1 at representative times
QPU scalability
vanilla Sampler vs Estimator res0
N=12..52
2(d-e) DFT spectrum
peaks vs eigenenergies
overlap explanation
Noisy simulator qualification
raw, post-selection, MPF
shot cost + acceptance
MPF validity
ε_A, ε_state
e(r)/e(2r) ≈ 4
fair Trotter baseline
candidate evidence
4-stage decomposition
exact -> noiseless Trotter
-> noisy raw -> post-selected
Round 2 QPU
Estimator res 0/1/2 + Sampler post-selection
N=12